

## Assignment 2

### MIN-HASHING AND LSH

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#### Question 1: Create k-Grams

**A:** Approximate Jaccard similarity between the pair of documents D1 and D2

```
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Exact 3-gram Jaccard(D1, D2) = 0.978

t = 20
Approx Jaccard = 0.9500
Time taken = 0.0054 seconds

t = 60
Approx Jaccard = 0.9833
Time taken = 0.0117 seconds

t = 150
Approx Jaccard = 0.9800
Time taken = 0.0223 seconds

t = 300
Approx Jaccard = 0.9600
Time taken = 0.0330 seconds

t = 600
Approx Jaccard = 0.9883
Time taken = 0.0560 seconds
```

**B:** Jaccard similarity between all pairs of documents for each type of k-gram

```
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Character 2-gram
D1 - D2 : 0.9811
D1 - D3 : 0.8157
D1 - D4 : 0.6444
D2 - D3 : 0.8000
D2 - D4 : 0.6413
D3 - D4 : 0.6530

Character 3-gram
D1 - D2 : 0.9780
D1 - D3 : 0.5804
D1 - D4 : 0.3051
D2 - D3 : 0.5680
D2 - D4 : 0.3059
D3 - D4 : 0.3121

Word 2-gram
D1 - D2 : 0.9408
D1 - D3 : 0.1823
D1 - D4 : 0.0302
D2 - D3 : 0.1737
D2 - D4 : 0.0303
D3 - D4 : 0.0161
```

## Question 2: Min-Hashing

A: Approximate Jaccard similarity between the pair of documents D1 and D2

```
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t = 20
Approximate Jaccard = 0.900000
Time taken = 0.005383 seconds

t = 60
Approximate Jaccard = 0.966667
Time taken = 0.012823 seconds

t = 150
Approximate Jaccard = 0.993333
Time taken = 0.020339 seconds

t = 300
Approximate Jaccard = 0.976667
Time taken = 0.031480 seconds

t = 600
Approximate Jaccard = 0.978333
Time taken = 0.054593 seconds

t = 800
Approximate Jaccard = 0.978750
Time taken = 0.070960 seconds

t = 1000
Approximate Jaccard = 0.975000
Time taken = 0.131359 seconds

t = 1500
Approximate Jaccard = 0.984000
```

### B: Analysis on t-value

- Jaccard similarity between the two documents is **0.978**. As we increase the number of hash functions (**t**), the MinHash estimate becomes closer to this exact value.
- For small **t** values (like **t = 20 or 60**), the approximation is less stable, and the error is relatively higher.
- As **t** increases, the error steadily decreases, but the computation time also increases. For example, **t = 600** gives very low error but takes about **0.055 seconds**, which is much slower.
- At **t = 150**, the estimated similarity (**0.9667**) is already very close to the exact value, with a **small error ( $\approx 0.011$ )** and reasonable **computation time ( $\sim 0.013$  s)**.
- Therefore, **t  $\approx$  150** is a good practical choice, since it provides high accuracy while keeping the computation time efficient.

## Question 3: LSH

A: All possible combinations of **b** and **r**

```
LSH Detection Probabilities (for  $\tau = 0.7$ ):  
-----  
r = 1, b = 160  
Estimated Threshold  $\approx 0.0063$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 1.000000  
D1-D4 : 1.000000  
D2-D3 : 1.000000  
D2-D4 : 1.000000  
D3-D4 : 1.000000  
-----  
r = 2, b = 80  
Estimated Threshold  $\approx 0.1118$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 1.000000  
D1-D4 : 0.999597  
D2-D3 : 1.000000  
D2-D4 : 0.999614  
D3-D4 : 0.999725  
-----  
r = 4, b = 40  
Estimated Threshold  $\approx 0.3976$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.991904  
D1-D4 : 0.293930  
D2-D3 : 0.987698  
D2-D4 : 0.296585  
D3-D4 : 0.317129  
-----  
r = 5, b = 32  
Estimated Threshold  $\approx 0.5000$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.886888  
D1-D4 : 0.081202  
D2-D3 : 0.857860  
D2-D4 : 0.082252  
D3-D4 : 0.090568  
-----
```

```
r = 8, b = 20  
Estimated Threshold  $\approx 0.6877$   
>>> Close to  $\tau = 0.7$  <<<  
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.228224  
D1-D4 : 0.001500  
D2-D3 : 0.195880  
D2-D4 : 0.001532  
D3-D4 : 0.001800  
-----  
r = 10, b = 16  
Estimated Threshold  $\approx 0.7579$   
Pair Probabilities:  
D1-D2 : 1.000000  
D1-D3 : 0.067145  
D1-D4 : 0.000112  
D2-D3 : 0.054526  
D2-D4 : 0.000115  
D3-D4 : 0.000140  
-----  
r = 16, b = 10  
Estimated Threshold  $\approx 0.8660$   
Pair Probabilities:  
D1-D2 : 0.999994  
D1-D3 : 0.001655  
D1-D4 : 0.000000  
D2-D3 : 0.001175  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 20, b = 8  
Estimated Threshold  $\approx 0.9013$   
Pair Probabilities:  
D1-D2 : 0.999722  
D1-D3 : 0.000150  
D1-D4 : 0.000000  
D2-D3 : 0.000098  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 32, b = 5  
Estimated Threshold  $\approx 0.9509$   
Pair Probabilities:  
D1-D2 : 0.965632  
D1-D3 : 0.000000  
D1-D4 : 0.000000  
D2-D3 : 0.000000  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----  
r = 40, b = 4  
Estimated Threshold  $\approx 0.9659$   
Pair Probabilities:  
D1-D2 : 0.879138  
D1-D3 : 0.000000  
D1-D4 : 0.000000  
D2-D3 : 0.000000  
D2-D4 : 0.000000  
D3-D4 : 0.000000  
-----
```

```

-----
r = 80, b = 2
Estimated Threshold  $\approx 0.9914$ 
Pair Probabilities:
D1-D2 : 0.308461
D1-D3 : 0.000000
D1-D4 : 0.000000
D2-D3 : 0.000000
D2-D4 : 0.000000
D3-D4 : 0.000000
-----

r = 160, b = 1
Estimated Threshold  $\approx 1.0000$ 
Pair Probabilities:
D1-D2 : 0.028362
D1-D3 : 0.000000
D1-D4 : 0.000000
D2-D3 : 0.000000
D2-D4 : 0.000000
D3-D4 : 0.000000

```

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| b   | r   | b×r | s*     | s*-τ        |
|-----|-----|-----|--------|-------------|
| 1   | 160 | 160 | 0.0063 | 0.6937 BEST |
| 2   | 80  | 160 | 0.1118 | 0.5882 BEST |
| 4   | 40  | 160 | 0.3976 | 0.3024 BEST |
| 5   | 32  | 160 | 0.5000 | 0.2000 BEST |
| 8   | 20  | 160 | 0.6877 | 0.0123 BEST |
| 10  | 16  | 160 | 0.7579 | 0.0579      |
| 16  | 10  | 160 | 0.8660 | 0.1660      |
| 20  | 8   | 160 | 0.9013 | 0.2013      |
| 32  | 5   | 160 | 0.9509 | 0.2509      |
| 40  | 4   | 160 | 0.9659 | 0.2659      |
| 80  | 2   | 160 | 0.9914 | 0.2914      |
| 160 | 1   | 160 | 1.0000 | 0.3000      |

Best choice : b = 8 hashes/band, r = 20 bands  
 Inflection : s\* = 0.6877 (target τ = 0.7)  
 S-curve : f(s) = 1 - (1 - s^8)^20

➔ For  $\tau = 0.7$  and 'b=80 & r=2' combination, Jaccard similarity between all pairs of documents:

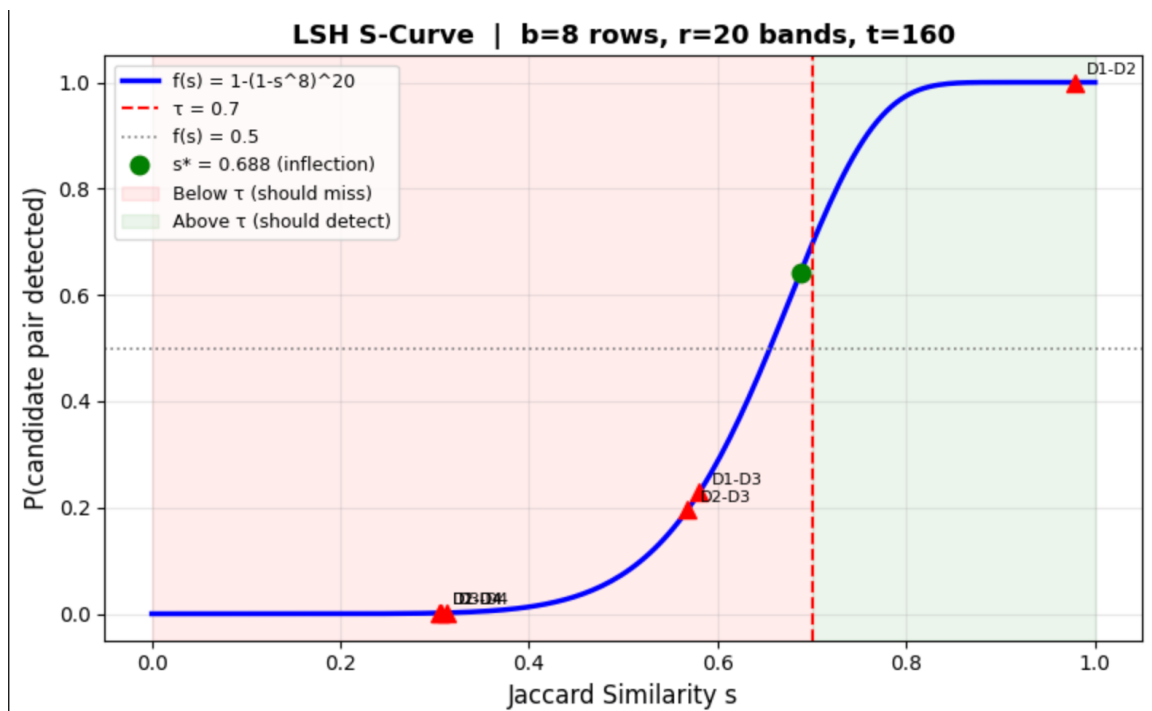
✓ Last Execution: 3:39:46 PM, Duration: 0.0s

Using: b = 8 hashes/band, r = 20 bands, t = 160  
 S-curve: f(s) = 1 - (1 - s^8)^20

| Pair  | Jaccard | f(J) = P(detected) | Above $\tau=0.7$ ? |
|-------|---------|--------------------|--------------------|
| D1-D2 | 0.9780  | 1.000000           | Yes                |
| D1-D3 | 0.5804  | 0.228224           | No                 |
| D1-D4 | 0.3051  | 0.001500           | No                 |
| D2-D3 | 0.5680  | 0.195880           | No                 |
| D2-D4 | 0.3059  | 0.001532           | No                 |
| D3-D4 | 0.3121  | 0.001800           | No                 |

## B: Conclusion (

- **Exact Similarities:**
  - D1–D2 = **0.978** -> clear near-duplicate
  - D1–D3, D2–D3  $\approx$  **0.58** -> moderate similarity
  - Pairs with D4  $\approx$  **0.30** -> low similarity
- **LSH Behaviour:**
  - **r=1, b=160** ( $t \approx 0.006$ ) -> All pairs detected ( $P \approx 1$ ) -> High false positives
  - **r=8, b=20** ( $t \approx 0.688 \approx \tau = 0.7$ )
    - D1–D2:  $P \approx 1$
    - Others:  $P \leq 0.23$   
-> Best separation
  - **r=160, b=1** ( $t \approx 1.0$ )
    - D1–D2:  $P \approx 0.028$   
-> High false negatives
- **Best configuration:** r=8, b=20
- Achieves a good balance between precision and recall
- Proper (r, b) tuning is critical for effective LSH performance



## Question 4: Min-Hashing on MovieLens dataset

➔ Load MovieLens dataset

```
Users   : 943
Movies  : 1682
Sample - User 5 rated 175 movies
```

➔ Exact Jaccard for all pairs

```
✓ Last Execution: 3:39:48 PM, Duration: 2.5s

Computing exact Jaccard for all 444,153 pairs...
Done in 2.59s
Pairs with  $J \geq 0.5$  : 10

Top 10 most similar pairs:
  User 408 & User 898 →  $J = 0.8387$ 
  User 328 & User 788 →  $J = 0.6730$ 
  User 489 & User 587 →  $J = 0.6299$ 
  User 600 & User 826 →  $J = 0.5455$ 
  User 451 & User 489 →  $J = 0.5333$ 
  User 674 & User 879 →  $J = 0.5217$ 
  User 554 & User 764 →  $J = 0.5170$ 
  User 197 & User 826 →  $J = 0.5130$ 
  User 197 & User 600 →  $J = 0.5000$ 
  User 800 & User 879 →  $J = 0.5000$ 
```

➔ MinHash for  $t = [50, 100, 200]$  with 5 runs each

```
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Exact similar pairs (ground truth) : 10

t= 50 | found=6.2 | FP=3.0 | FN=6.8 | time=0.8s
t= 100 | found=3.2 | FP=0.2 | FN=7.0 | time=1.4s
t= 200 | found=2.6 | FP=0.0 | FN=7.4 | time=2.6s
```

➔ For the false positives and negatives, report the averages for 5 different runs.

```
✓ Last Execution: 3:42:43 PM, Duration: 0.5s
```

| t   | Found | Avg FP | Avg FN | FP%   | FN%   | Time(s) |
|-----|-------|--------|--------|-------|-------|---------|
| 50  | 6.2   | 3.0    | 6.8    | 48.4% | 68.0% | 0.8s    |
| 100 | 3.2   | 0.2    | 7.0    | 6.2%  | 70.0% | 1.4s    |
| 200 | 2.6   | 0.0    | 7.4    | 0.0%  | 74.0% | 2.6s    |

## Question 5: LSH on MovieLens dataset

→ Candidate pairs with a similarity of at least 0.6.

Exact pairs with  $J \geq 0.6$ : 3

[t=50 r=5 b=10] Run 1 | candidates= 513 | found= 5 | FP= 2 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 2 | candidates= 599 | found= 5 | FP= 2 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 3 | candidates= 627 | found= 4 | FP= 2 | FN= 1 | 0.2s

[t=50 r=5 b=10] Run 4 | candidates= 732 | found= 6 | FP= 3 | FN= 0 | 0.2s

[t=50 r=5 b=10] Run 5 | candidates= 469 | found= 3 | FP= 1 | FN= 1 | 0.2s

— AVG: FP=2.0 | FN=0.4 | Found=4.6

[t=100 r=5 b=20] Run 1 | candidates=1348 | found= 2 | FP= 1 | FN= 2 | 0.4s

[t=100 r=5 b=20] Run 2 | candidates= 901 | found= 2 | FP= 0 | FN= 1 | 0.4s

[t=100 r=5 b=20] Run 3 | candidates= 978 | found= 4 | FP= 1 | FN= 0 | 0.4s

[t=100 r=5 b=20] Run 4 | candidates= 945 | found= 3 | FP= 0 | FN= 0 | 0.4s

[t=100 r=5 b=20] Run 5 | candidates=1028 | found= 4 | FP= 1 | FN= 0 | 0.4s

— AVG: FP=0.6 | FN=0.6 | Found=3.0

[t=200 r=5 b=40] Run 1 | candidates=2449 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 2 | candidates=2202 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 3 | candidates=1803 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 4 | candidates=2496 | found= 3 | FP= 0 | FN= 0 | 0.8s

[t=200 r=5 b=40] Run 5 | candidates=1707 | found= 3 | FP= 0 | FN= 0 | 0.8s

— AVG: FP=0.0 | FN=0.0 | Found=3.0

[t=200 r=10 b=20] Run 1 | candidates= 6 | found= 1 | FP= 0 | FN= 2 | 0.8s

[t=200 r=10 b=20] Run 2 | candidates= 8 | found= 1 | FP= 0 | FN= 2 | 0.8s

[t=200 r=10 b=20] Run 3 | candidates= 2 | found= 1 | FP= 0 | FN= 2 | 0.7s

[t=200 r=10 b=20] Run 4 | candidates= 4 | found= 1 | FP= 0 | FN= 2 | 0.7s

[t=200 r=10 b=20] Run 5 | candidates= 6 | found= 2 | FP= 0 | FN= 1 | 0.7s

— AVG: FP=0.0 | FN=1.8 | Found=1.2

→ The number of false positives and false negatives, taking the average over 5 runs.

| $\tau = 0.6$   Exact pairs = 3 |       |        |        |       |       |      |
|--------------------------------|-------|--------|--------|-------|-------|------|
| Config                         | Found | Avg FP | Avg FN | FP%   | FN%   | Time |
| t=50 r=5 b=10                  | 4.6   | 2.0    | 0.4    | 43.5% | 13.3% | 0.2s |
| t=100 r=5 b=20                 | 3.0   | 0.6    | 0.6    | 20.0% | 20.0% | 0.4s |
| t=200 r=5 b=40                 | 3.0   | 0.0    | 0.0    | 0.0%  | 0.0%  | 0.8s |
| t=200 r=10 b=20                | 1.2   | 0.0    | 1.8    | 0.0%  | 60.0% | 0.8s |

➔ Candidate pairs with a similarity of at least 0.8.

Exact pairs with  $J \geq 0.8$ : 1

[t=50 r=5 b=10] Run 1 | candidates= 574 | found= 1 | FP= 0 | FN= 0 | 0.2s  
[t=50 r=5 b=10] Run 2 | candidates= 347 | found= 1 | FP= 0 | FN= 0 | 0.2s  
[t=50 r=5 b=10] Run 3 | candidates= 604 | found= 1 | FP= 0 | FN= 0 | 0.2s  
[t=50 r=5 b=10] Run 4 | candidates= 492 | found= 1 | FP= 0 | FN= 0 | 0.2s  
[t=50 r=5 b=10] Run 5 | candidates= 453 | found= 1 | FP= 0 | FN= 0 | 0.2s  
— AVG: FP=0.0 | FN=0.0 | Found=1.0

[t=100 r=5 b=20] Run 1 | candidates= 876 | found= 0 | FP= 0 | FN= 1 | 0.4s  
[t=100 r=5 b=20] Run 2 | candidates= 760 | found= 0 | FP= 0 | FN= 1 | 0.4s  
[t=100 r=5 b=20] Run 3 | candidates=1273 | found= 1 | FP= 0 | FN= 0 | 0.4s  
[t=100 r=5 b=20] Run 4 | candidates=1878 | found= 1 | FP= 0 | FN= 0 | 0.4s  
[t=100 r=5 b=20] Run 5 | candidates=1343 | found= 1 | FP= 0 | FN= 0 | 0.4s  
— AVG: FP=0.0 | FN=0.4 | Found=0.6

[t=200 r=5 b=40] Run 1 | candidates=1993 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=5 b=40] Run 2 | candidates=1736 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=5 b=40] Run 3 | candidates=2554 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=5 b=40] Run 4 | candidates=1978 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=5 b=40] Run 5 | candidates=2544 | found= 1 | FP= 0 | FN= 0 | 0.8s  
— AVG: FP=0.0 | FN=0.0 | Found=1.0

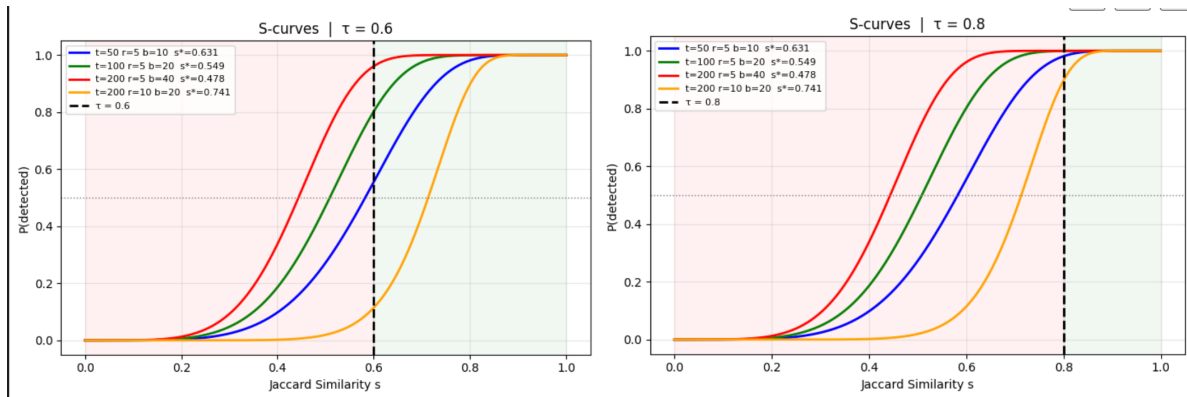
[t=200 r=10 b=20] Run 1 | candidates= 12 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=10 b=20] Run 2 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=10 b=20] Run 3 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=10 b=20] Run 4 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s  
[t=200 r=10 b=20] Run 5 | candidates= 5 | found= 1 | FP= 0 | FN= 0 | 0.8s  
— AVG: FP=0.0 | FN=0.0 | Found=1.0

➔ The number of false positives and false negatives, taking the average over 5 runs.

| $\tau = 0.8$   Exact pairs = 1 |       |        |        |      |       |      |
|--------------------------------|-------|--------|--------|------|-------|------|
| Config                         | Found | Avg FP | Avg FN | FP%  | FN%   | Time |
| t=50 r=5 b=10                  | 1.0   | 0.0    | 0.0    | 0.0% | 0.0%  | 0.2s |
| t=100 r=5 b=20                 | 0.6   | 0.0    | 0.4    | 0.0% | 40.0% | 0.4s |
| t=200 r=5 b=40                 | 1.0   | 0.0    | 0.0    | 0.0% | 0.0%  | 0.8s |
| t=200 r=10 b=20                | 1.0   | 0.0    | 0.0    | 0.0% | 0.0%  | 0.8s |



## → LSH S-curves for all configs



## → Analysis

- $\tau = 0.6$ : Using more hash functions generally improved accuracy. The configuration **( $t=200, r=5, b=40$ )** performed best with **0% false positives and 0% false negatives**.
- The combination **( $t=50, r=5, b=10$ )** produced many **false positives (~43.5%)**, meaning it detected many pairs that were not actually similar.
- **( $t=200, r=10, b=20$ )** reduced false positives but introduced **high false negatives**, missing several truly similar pairs.
- $\tau = 0.8$ : Since the similarity requirement is stricter, fewer true pairs exist. Most configurations had **0% false positives**, but **( $t=100, r=5, b=20$ )** missed some valid pairs (**~40% false negatives**).
- Overall, higher hash counts with balanced banding (e.g.,  **$t=200, r=5, b=40$** ) gave the most reliable results.

**GitHub link:**

[https://github.com/jahanvi0106/ML\\_with\\_BigData/tree/Assignment2](https://github.com/jahanvi0106/ML_with_BigData/tree/Assignment2)